Sequence-Level Determinants of Cholesterol Specificity in Lipid-Binding Proteins: An Interpretable, Homology-Aware Comparative Analysis

Author One[1], Author Two[1], Author Three[2]

[1]Affiliation Placeholder, Department Placeholder, Institution Placeholder, City, Country

[2]Affiliation Placeholder, Department Placeholder, Institution Placeholder, City, Country

Corresponding author: corresponding.author@placeholder.edu

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1 Introduction

Lipid-binding proteins are central to membrane organization, transport, signaling, metabolism, and cellular communication. They include integral membrane proteins, peripheral membrane proteins, receptors, transporters, enzymes, and scaffold proteins whose activity depends on direct or indirect interaction with membrane-associated small molecules. Although many studies have focused on particular lipid classes or individual receptors, the broader question of whether proteins that bind one membrane lipid class can be distinguished from proteins that bind another remains incompletely resolved. Addressing this question is important both for understanding membrane biology and for developing generalizable computational frameworks for protein function inference.

Among membrane lipids, cholesterol occupies a particularly distinctive position. In animal membranes, cholesterol strongly influences bilayer thickness, order, packing, permeability, phase behavior, and membrane-protein organization. Cholesterol is not simply another hydrophobic ligand; it has a rigid tetracyclic sterol ring system, a small polar hydroxyl headgroup, and a short alkyl tail that together generate a characteristic combination of shape, polarity, and membrane-partitioning behavior. These properties allow cholesterol to modulate membrane-protein conformational equilibria, stabilize selected folds, influence oligomerization, and alter the microenvironment of receptors, channels, and transporters. Consequently, a protein that binds cholesterol may require a different sequence and physicochemical context from a protein that binds phospholipids, phosphoinositides, fatty acids, or other membrane-associated ligands.

Distinguishing cholesterol-binding proteins from other lipid-binding proteins is therefore a specificity problem rather than a generic ligand-binding problem. A model trained to separate cholesterol-binding proteins from all remaining proteins could achieve strong nominal accuracy while learning features that primarily reflect membrane localization, taxonomic composition, or protein-family membership. Such a model would not answer the more biologically relevant question: what, if anything, distinguishes cholesterol-binding proteins from proteins that already bind other biologically meaningful lipids? To address this question rigorously, the comparison class must be composed of non-cholesterol lipid-binding proteins rather than arbitrary non-binders.

Previous work on cholesterol recognition has often emphasized sequence motifs such as CRAC and CARC, as well as specific residue interactions observed in selected structures. These approaches have been valuable for hypothesis generation, but they face important limitations. Motif-only analyses can overstate the specificity of short patterns that occur frequently by chance in hydrophobic or transmembrane sequences. Conversely, highly flexible black-box predictors may achieve predictive performance without revealing the mechanistic or physicochemical basis of the separation. In the present context, neither extreme is sufficient: a simple motif scan risks missing distributed sequence organization, whereas an opaque model risks obscuring the biological interpretation.

These considerations motivate a sequence-level statistical analysis centered on interpretable features. Such an approach enables direct testing of whether cholesterol-binding proteins differ from

other lipid-binding proteins in residue composition, hydrophobicity, polarity, charge balance, aromaticity, local hydrophobic organization, and simple motif-like patterns. This is especially valuable for a first comparative study because the resulting signal can be examined feature by feature and interpreted in relation to membrane insertion, lipid-facing surfaces, aromatic anchoring, and head-group recognition. Position-normalized profiles further allow the analysis to move beyond overall composition and test whether physicochemical differences are distributed differently along protein sequences.

At the same time, any sequence-based comparative study of structural interaction data must confront homology. Protein structures are highly redundant, certain families are disproportionately represented in the Protein Data Bank, and closely related homologs can inflate both statistical significance and predictive accuracy if they appear in both training and test sets. This issue is especially acute for membrane proteins, where certain receptor and transporter families contribute many related structures. For that reason, a central design principle of the present study was homology-aware validation. We therefore combined exact-sequence deduplication, removal of proteins observed in both lipid classes, sequence clustering at 80% identity, mixed-label cluster removal, and a final cluster-aware cross-validation analysis to determine how much of the observed signal persists when residual family overlap is prevented.

The goal of this work was not to produce a maximal-accuracy predictor in the absence of interpretability. Rather, our aim was to build a reproducible and biologically careful framework for testing whether cholesterol-binding proteins are distinguishable from other lipid-binding proteins at the sequence level, and, if so, to identify the specific sequence-level properties that contribute to that distinction. We show that cholesterol-binding proteins display coherent differences in global composition, local membrane-like sequence organization, and cluster-aware predictive signal, supporting the view that cholesterol specificity is associated with distributed sequence-level adaptation rather than a single universal motif.

2 Methods

2.1 Study design

The study was designed to answer a narrowly defined biological question: what distinguishes cholesterol-binding proteins from other lipid-binding proteins? The primary aim was to identify sequence-level properties associated with cholesterol specificity, not merely to separate membrane proteins from soluble proteins or lipid binders from non-binders. Accordingly, the positive and comparison classes were defined using direct structural interaction evidence with carefully curated ligand classes.

The positive class comprised proteins with direct evidence of interaction with the cholesterol ligand `CLR`. The comparison class comprised proteins with direct evidence of interaction with a curated

set of non-sterol lipids. Proteins that had evidence of interaction with both cholesterol and the curated non-cholesterol lipid set were removed from the primary binary analysis. This decision was crucial: such proteins are biologically interesting, but they are not clean negatives for a cholesterol-specificity comparison. Keeping them in the binary design would have introduced label conflict and diluted the interpretability of subsequent statistical comparisons.

The observation level during dataset construction was the protein–ligand interaction record associated with a structural entry. However, the final analytical unit was a protein-level sequence record. Observation-level data are useful for provenance and metadata retention, but they are not an appropriate final unit for inferential statistics because the same protein may appear in multiple structures, constructs, ligand-bound states, and experimental contexts. Thus, the workflow was explicitly designed to separate provenance-preserving observation-level assembly from protein-level and homology-reduced sequence-level analysis.

Inclusion rules required a protein polymer, a curated ligand identity, a usable protein sequence, and sufficient annotation to support class assignment. Exclusion rules removed ambiguous ligands, detergent-like compounds, crystallization additives, extremely short proteins, invalid sequences, and proteins participating in both cholesterol and non-cholesterol classes. These decisions were made to keep the binary comparison centered on specificity rather than generic hydrophobic association.

2.2 Data sources

BioLiP was used as the primary interaction source because it is curated for biologically relevant ligand–protein interactions rather than simple co-occurrence of ligands within a structural entry [7]. This distinction was important for the present study because the biological question depends on direct protein–ligand association rather than mere presence of a lipid somewhere in a multicomponent structure.

The RCSB Protein Data Bank was used as the structural metadata and identifier-enrichment layer [5]. RCSB provides entry-level and entity-level annotations, including experimental method, structural context, and standardized ligand identifiers. Although BioLiP supplied the interaction evidence used for primary labeling, RCSB served as the authoritative structural archive and the reference layer for ligand identifiers such as `CLR`.

ChEBI was used as an ontology support layer for ligand curation [2]. The comparison class in this study required more than a list of three-letter ligand codes; it required a biologically coherent distinction between cholesterol, non-cholesterol lipids, detergents, additives, and ambiguous compounds. ChEBI was therefore used to support manual ligand-family review, especially in borderline cases involving sterol-adjacent molecules or noncanonical lipids.

UniProt was used as the protein reference layer [6]. UniProt accessions provided a stable biological identifier when available and assisted interpretation, duplicate handling, and provenance tracking. However, because not all structural sequences map cleanly to a single accession and because con-

structs or engineered proteins may differ from canonical sequences, exact sequence identity remained the principal unit for collapse during cleaning.

2.3 Ligand curation

The ligand curation process began with the cholesterol ligand `CLR`, which served as the anchor of the positive class. `CLR` was chosen because it is a direct and operationally clean representation of cholesterol in PDB-centered structural data. We intentionally did not begin with a broad sterol family because that would have blurred the specificity question at the outset.

Non-cholesterol lipid selection proceeded by reviewing BioLiP ligand frequencies and names, then filtering candidate ligands according to biological relevance, support size, and chemical interpretability. The final curated non-cholesterol panel included the following ligands: `PGW`, `CDL`, `LBN`, `CPL`, `T7X`, `PT5`, `EIC`, `PLM`, `ACD`, and `MYR`. These ligands represent phosphatidylglycerol, cardiolipin, phosphatidylcholine species, phosphatidylinositol species, and biologically relevant fatty acids. Together they provide a comparison set that is chemically meaningful, sufficiently large, and clearly distinct from cholesterol.

Several classes of ligands were explicitly excluded. First, sterol-adjacent ligands such as cholesterol analogs and cholesterol-derived compounds were removed because they would weaken the intended specificity contrast between cholesterol and non-cholesterol lipid binding. Second, detergents and additives were excluded because they frequently appear in membrane-protein structures but are not reliable indicators of biologically meaningful lipid recognition. Third, synthetic short-chain phospholipids were excluded because they can behave differently from native membrane lipids and may reflect experimental convenience more than physiological relevance. Fourth, low-support sphingolipids were excluded from the primary comparison set because the number of available protein interactions was too small to support a stable and interpretable binary analysis. This curation strategy favored biological coherence over maximal inclusion.

2.4 Dataset construction

Raw observations were assembled from BioLiP interaction annotations. Each raw row retained the interacting ligand identifier, label assignment, protein sequence, UniProt identifier when available, protein chain or entity provenance, PDB entry identifiers, ligand types observed, and associated structural metadata. The raw observation table was intentionally broad because the first objective was to preserve provenance and support later review of attrition and exclusion decisions.

Useful metadata fields retained during the raw stage included protein sequence, UniProt identifier, ligand identifier, ligand family, PDB identifiers, count of associated observations, and cluster provenance where applicable. Experimental fields from structural metadata were also retained during earlier stages of the workflow because they are helpful for later quality review and interpretation.

However, these observation-level records were not used directly as the final analytical unit because multiple structures of the same protein would otherwise produce pseudo-replication.

2.5 Cleaning and deduplication

Cleaning began by converting the observation-level dataset to a protein-level representation. Exact sequence was treated as the initial biological unit because it provides an unambiguous operational definition that does not depend on perfect external accession mapping. UniProt identifiers were used when available as a stable biological reference, but because UniProt mappings can be absent, multiple, or construct-dependent, exact sequence identity remained primary. When no unique accession was available, a sequence-hash identifier was used as a stable fallback.

Duplicate observation collapse merged repeated occurrences of the same exact sequence while preserving aggregate metadata, including the number of observations, the set of ligands observed, and the contributing PDB identifiers. This step was statistically necessary because repeated structural observations of the same or identical sequences would otherwise inflate the apparent sample size. It was also biologically appropriate because the study sought to compare protein sequences, not structure deposition frequency.

Quality filtering then removed proteins shorter than 50 amino acids, along with any invalid sequences containing nonstandard residue characters. The short-sequence filter was introduced to remove likely fragments, highly truncated constructs, or minimal peptides that would be difficult to compare meaningfully against full-length or domain-level proteins. Invalid-sequence filtering ensured that downstream physicochemical feature calculations and profile interpolation were well defined. In practice, no invalid sequences were observed after the earlier curation stages.

Proteins observed in both cholesterol and other-lipid classes were then removed. This step was especially important. Such proteins may represent genuine multi-lipid binders, but they do not fit cleanly into a binary specificity framework. Because the study question asked what distinguishes cholesterol-binding proteins from proteins binding other lipids, the presence of a protein in both classes would create label ambiguity and weaken both statistical comparison and classifier interpretation.

2.6 Homology reduction

To reduce residual homology leakage, cleaned protein sequences were clustered using CD-HIT at 80% sequence identity [4]. The 80% threshold was chosen as a pragmatic compromise. It is strict enough to collapse clearly similar homologs, closely related isoforms, and many family-level near-redundancies that would otherwise inflate performance, while still retaining sufficient diversity for meaningful comparison between the cholesterol and non-cholesterol classes.

After clustering, clusters containing proteins from both classes were removed in their entirety.

This mixed-label cluster removal step was necessary because a mixed cluster indicates that highly similar proteins are present in both classes. Allowing one member of such a cluster into training and another into test would undermine the goal of homology-aware evaluation and complicate interpretation of class-specific features. For the remaining clusters, one representative protein was retained. This representative-based reduction provided a homology-reduced dataset appropriate for feature construction, exploratory statistics, and initial model development.

2.7 Feature construction

Baseline features were designed to remain interpretable. Twenty amino-acid composition features were computed as fractions of each standard amino acid in the sequence. Additional global features included sequence length, hydrophobic residue fraction, positively charged residue fraction, negatively charged residue fraction, aromatic residue fraction, polar residue fraction, nonpolar residue fraction, average hydrophobicity, average charge score, and average polarity.

Hydrophobicity was computed using the Kyte–Doolittle scale. Charge was encoded with a simple residue-level score in which lysine, arginine, and histidine were assigned $+1$, aspartate and glutamate were assigned -1, and all other residues were assigned 0. Polarity was summarized using the Grantham polarity scale. These mappings were selected because they are transparent, widely understood, and suitable for interpretation in a membrane-protein context.

In addition to global features, three position-dependent profile feature sets were constructed: hydrophobicity, charge, and polarity profiles. For each sequence, residue-wise property values were first assigned according to the mappings above. The sequence was then mapped onto a normalized positional domain $[0, 1]$, and each profile was linearly interpolated to 100 equally spaced positions. This procedure enabled comparison across proteins of different lengths without naive padding and allowed the analysis to ask whether physicochemical properties were distributed differently along the relative sequence coordinate.

2.8 Exploratory statistical analysis

Exploratory statistical analyses were performed on the homology-reduced representative dataset. For baseline features, group-specific means and standard deviations were computed for cholesterol-binding and other-lipid-binding proteins. Group differences were then assessed with nonparametric two-sided Mann–Whitney tests, and false discovery rate correction was applied using the Benjamini–Hochberg procedure. These analyses were designed to identify which interpretable sequence properties differed systematically between classes.

Principal component analysis (PCA) was then applied to the baseline feature matrix after standardization. PCA was used not as a classifier, but as an exploratory dimensionality-reduction technique to visualize global structure in the feature space and to identify the leading axes of variation and

their dominant contributing features.

For the profile features, mean hydrophobicity, charge, and polarity curves were computed separately for each class over the normalized sequence domain. Difference curves were then constructed by subtracting the non-cholesterol class mean from the cholesterol class mean. These analyses were exploratory and interpretation-focused; they were intended to reveal whether sequence organization differed in a position-dependent manner beyond global composition averages.

2.9 Classification analysis

Classification analyses were included as supporting evidence rather than as the primary goal of the study. Two model classes were used: logistic regression and random forest. Logistic regression was chosen because it provides a linear and interpretable baseline, whereas random forest provides a nonlinear comparison that remains comparatively interpretable through feature-importance analysis.

Models were evaluated under stratified five-fold cross-validation. Feature sets included baseline features alone, profile-only feature sets, combined baseline-plus-profile feature sets, and an ablation model in which hydrophobic features were removed from the combined feature set. The ablation analysis was included to test whether performance was driven entirely by hydrophobicity-related information or whether additional sequence properties contributed independently.

2.10 Cluster-aware validation

Ordinary cross-validation can overestimate performance when homologous proteins are split across training and test folds. To address this problem, a stricter validation scheme was implemented using `StratifiedGroupKFold`, with homology clusters derived from the 80% CD-HIT reduction used as the group variable. Under this scheme, all proteins from the same cluster were confined to the same fold. This prevents family-level near-redundancy from leaking between training and test sets and provides a more realistic estimate of out-of-family generalization.

Cluster-aware validation was applied to the baseline and combined feature models. Performance differences between ordinary and cluster-aware validation were interpreted as evidence of the degree to which homologous similarity inflated standard cross-validation estimates.

2.11 Local motif and local-pattern analysis

As a focused follow-up, we performed a local motif and local-pattern analysis to test whether cholesterol-binding proteins were enriched for simple cholesterol-related or membrane-interface-like sequence patterns. This analysis was explicitly framed as heuristic rather than mechanistic.

CRAC-like motifs were defined using the rule [L/V]–X(1–5)–Y–X(1–5)–[R/K], consistent with a

permissive cholesterol recognition amino acid consensus pattern [3]. CARC-like motifs were defined using the reverse-oriented rule [R/K]–X(1–5)–[Y/F/W]–X(1–5)–[L/V], following the general CARC concept [1]. In addition, we computed cysteine-cluster indicators for `CC`, `CXC`, and `CXXC`; the longest contiguous hydrophobic run; the presence and count of aromatic-enriched windows; and the maximum local hydrophobic window score. Aromatic-enriched windows were defined as sliding windows of length 11 with aromatic fraction at least 0.30. The maximum local hydrophobic window score was defined as the maximum fraction of hydrophobic residues within a sliding window of length 11.

These features were included because cholesterol-binding proteins appeared enriched for hydrophobic, aromatic, and cysteine-related signals in the global analysis. They provide a transparent way to test whether those global trends are reflected in simple local sequence organization. However, they were not treated as definitive cholesterol-binding motifs, because pattern enrichment alone does not prove a structural or mechanistic mode of recognition.

3 Results

3.1 Dataset construction, cleaning, and attrition

The study began with an observation-level dataset of 5,396 curated protein–ligand interaction records assembled after ligand curation and raw dataset construction. These raw observations represented 1,221 unique protein sequences. Quality control removed 70 short proteins shorter than 50 amino acids and no invalid-sequence rows. Exact duplicate structural observations were then collapsed, removing 4,129 redundant observation rows and yielding 1,197 unique sequences after quality filtering and sequence-level collapse.

To maintain a clean binary specificity comparison, proteins observed with both cholesterol and curated non-cholesterol lipids were removed. This step eliminated 48 multi-lipid sequences corresponding to 364 observation rows. The resulting cleaned protein-level dataset contained 1,149 proteins, including 695 cholesterol-binding proteins and 454 other-lipid-binding proteins. This cleaning stage converted the project from a provenance-rich structural interaction table to an inferentially meaningful protein-level sequence dataset.

Homology reduction was then applied using CD-HIT at 80% identity. The cleaned dataset of 1,149 proteins was organized into 493 clusters. Ten mixed-label clusters contained both cholesterol and non-cholesterol proteins and were removed entirely, eliminating 62 proteins. After exclusion of mixed-label clusters, 1,087 proteins remained. One representative sequence was then retained per cluster, producing a final homology-reduced dataset of 483 proteins, comprising 275 cholesterol-binding proteins and 208 other-lipid-binding proteins. This homology-reduced set formed the basis of the primary exploratory analyses and initial model-development stage.

Table 1: Dataset construction, cleaning, and homology-reduction attrition summary.

Processing stage	**Count**
Raw rows	5,396
Raw unique sequences	1,221
Short rows removed (< 50 aa)	70
Invalid-sequence rows removed	0
Exact duplicate observation rows removed during collapse	4,129
Unique sequences after quality filtering and collapse	1,197
Multi-lipid sequences removed	48
Multi-lipid observations removed	364
Final cleaned proteins	1,149
Input cleaned proteins to CD-HIT	1,149
CD-HIT identity threshold	0.80
Total clusters	493
Mixed-label clusters removed	10
Proteins removed because of mixed-label clusters	62
Proteins remaining after cluster filter	1,087
Removed by within-cluster representative selection	604
Final homology-reduced representatives	483

Table 2: Final class composition after cleaning and after homology reduction.

Dataset stage	**Cholesterol**	**Other lipid**
Cleaned protein-level dataset	695	454
Homology-reduced representative dataset	275	208

3.2 Baseline compositional differences

Comparison of the baseline interpretable features revealed clear and coherent differences between cholesterol-binding proteins and proteins that bound other curated lipids. The cholesterol-binding class was substantially more hydrophobic by multiple measures. The mean hydrophobic residue fraction was 0.533 in cholesterol-binding proteins versus 0.486 in the other-lipid class (false discovery rate, FDR, 1.31×10^{-12}), and mean Kyte–Doolittle hydrophobicity was 0.396 versus 0.111 (FDR 1.65×10^{-11}). Conversely, cholesterol-binding proteins were less polar and less acidic, with a mean Grantham polarity of 7.84 versus 8.11 (FDR 5.05×10^{-12}) and a negative-residue fraction of 0.075 versus 0.092 (FDR 2.55×10^{-6}).

Residue-specific composition differences were also pronounced. Cholesterol-binding proteins had a

substantially higher cysteine fraction (0.0273 versus 0.0153, FDR 4.77×10^{-23}), a higher leucine fraction (0.1256 versus 0.1098, FDR 6.00×10^{-8}), and a higher phenylalanine fraction (0.0657 versus 0.0579, FDR 2.59×10^{-5}). In contrast, the non-cholesterol class showed greater enrichment of acidic residues and a higher glycine fraction. These results indicate that cholesterol-binding proteins are distinguished not only by a generic increase in hydrophobicity, but also by a coordinated shift toward a less polar, less acidic, and more membrane-adapted residue composition.

Biologically, these differences are consistent with proteins that spend more sequence real estate in membrane-embedded or membrane-interface environments. Enrichment of hydrophobic, aromatic, and selected sulfur-containing residues may reflect the need to stabilize transmembrane or interfacial regions capable of supporting sterol interaction. At the same time, the lower abundance of acidic residues suggests a relative depletion of strongly polar or solvent-exposed sequence segments in the cholesterol-binding class.

Table 3: Key baseline feature differences between cholesterol-binding and other-lipid-binding proteins in the homology-reduced dataset.

Feature	**Cholesterol mean**	**Other-lipid mean**	**FDR**
`hydrophobic_residue_fraction`	0.533	0.486	1.31×10^{-12}
`average_hydrophobicity_kd`	0.396	0.111	1.65×10^{-11}
`average_polarity_grantham`	7.84	8.11	5.05×10^{-12}
`negative_residue_fraction`	0.075	0.092	2.55×10^{-6}
`aa_frac_C`	0.0273	0.0153	4.77×10^{-23}
`aa_frac_L`	0.1256	0.1098	6.00×10^{-8}
`aa_frac_F`	0.0657	0.0579	2.59×10^{-5}

3.3 Principal component analysis of baseline features

PCA of the standardized baseline feature matrix showed that the first two principal components together explained 40.8% of the total variance. PC1 alone explained 31.2%, whereas PC2 explained 9.6%. Inspection of the loadings indicated that PC1 was primarily a hydrophobicity–polarity axis, dominated by average polarity, average hydrophobicity, hydrophobic residue fraction, polar residue fraction, nonpolar residue fraction, and acidic residue content. PC2 was driven more strongly by charge and aromatic composition, including average charge score, arginine fraction, positive residue fraction, histidine fraction, tryptophan fraction, and aromatic residue fraction.

These loading patterns agree closely with the univariate feature analyses. The leading axis of variation in the data separates sequences along a membrane-adaptation gradient, whereas the second axis appears to capture subtler differences in charged and aromatic composition. The PCA scatter plot should therefore be interpreted primarily as a low-dimensional view of the same

broad biochemical distinctions already identified feature by feature rather than as an independent discovery.

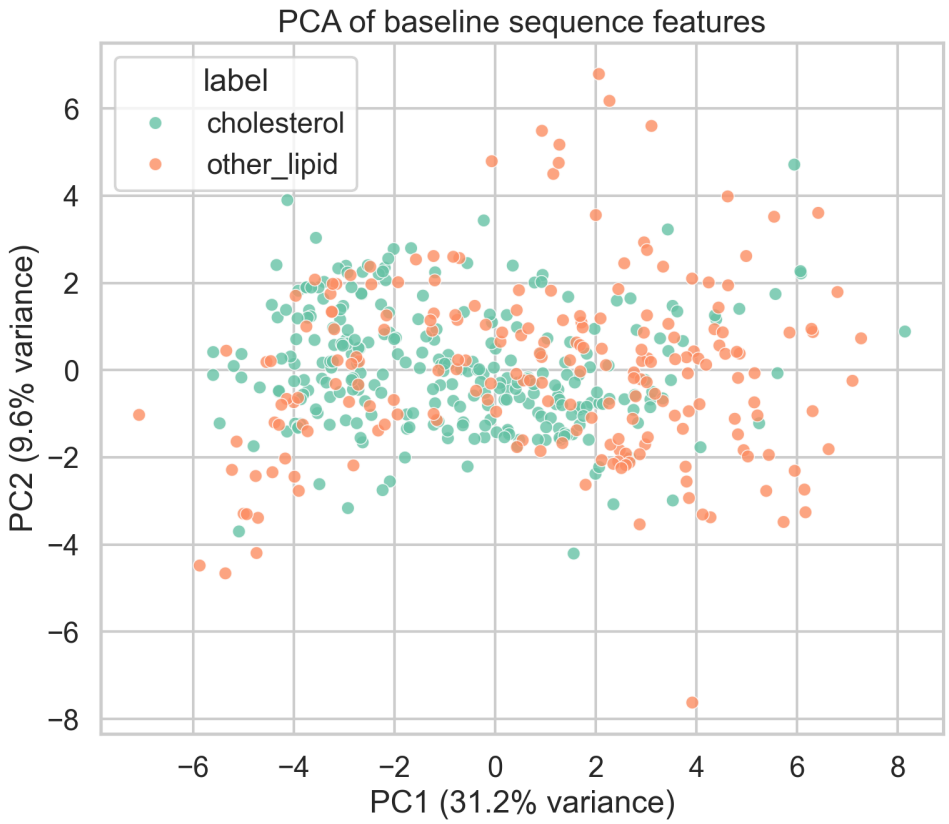

Figure 1: Principal component analysis of the baseline feature matrix. PC1 captured a dominant hydrophobicity/polarity axis, whereas PC2 reflected charge- and aromaticity-related variation.

3.4 Position-normalized profile analysis

Position-normalized physicochemical profiles demonstrated that cholesterol-binding proteins differed from the comparison class not only in overall composition but also in the spatial arrangement of sequence properties along the normalized sequence coordinate. The hydrophobicity profile showed its strongest difference near normalized position ~ 0.62, where cholesterol-binding proteins exhibited substantially higher mean hydrophobicity than proteins binding other lipids. The charge profile showed its largest class difference near normalized position ~ 0.31, with a relative shift toward more positive local values in the cholesterol-binding class. The polarity profile showed its strongest class difference near normalized position ~ 0.90, where the other-lipid class was more polar.

These results suggest that the distinguishing signal is not exhausted by global averages. Instead, cholesterol-binding proteins appear to differ in how hydrophobic, charged, and polar segments

are arranged across their sequence length after normalization. Such positional organization may reflect differences in transmembrane segment packing, interfacial topology, local headgroup-contact regions, or repeated membrane-associated modules.

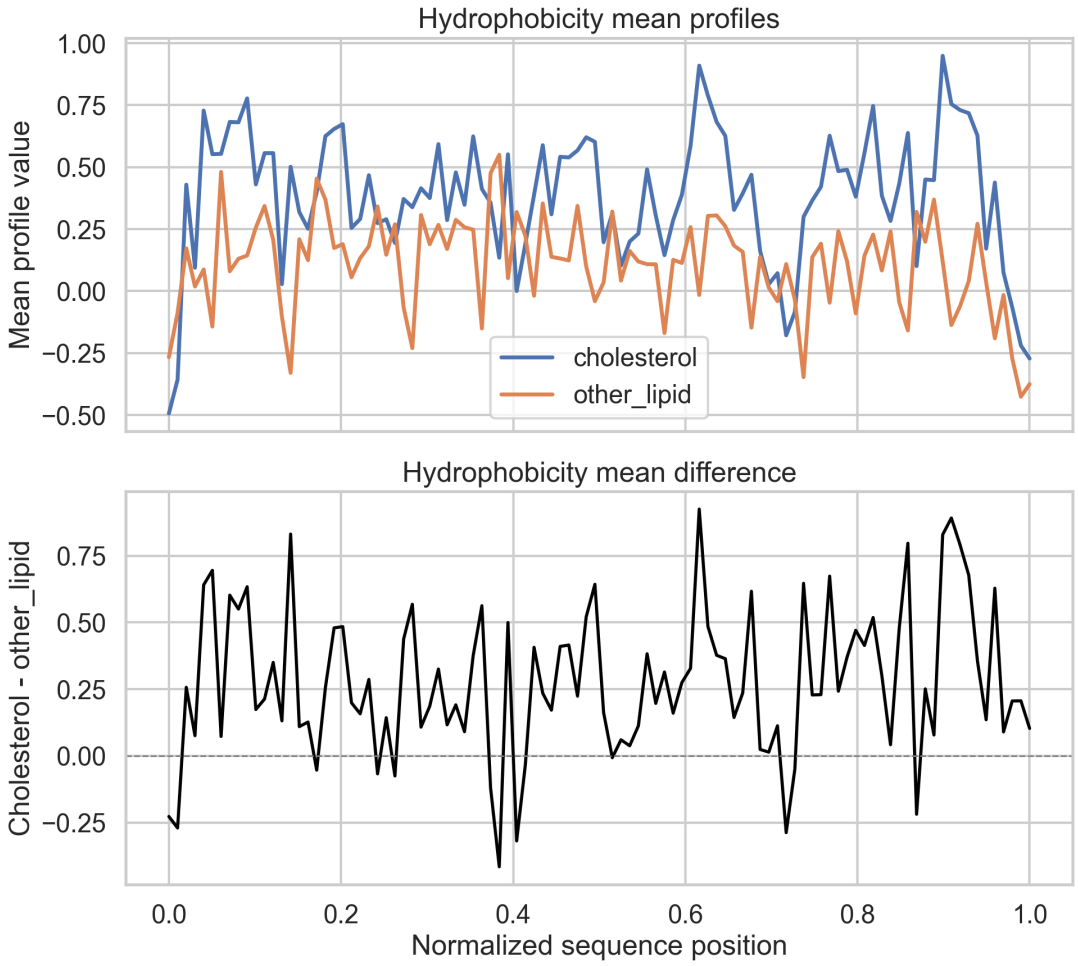

Figure 2: Position-normalized hydrophobicity profiles for cholesterol-binding and other-lipid-binding proteins. The largest difference occurred near normalized position ~ 0.62, consistent with stronger local hydrophobic organization in the cholesterol-binding class.

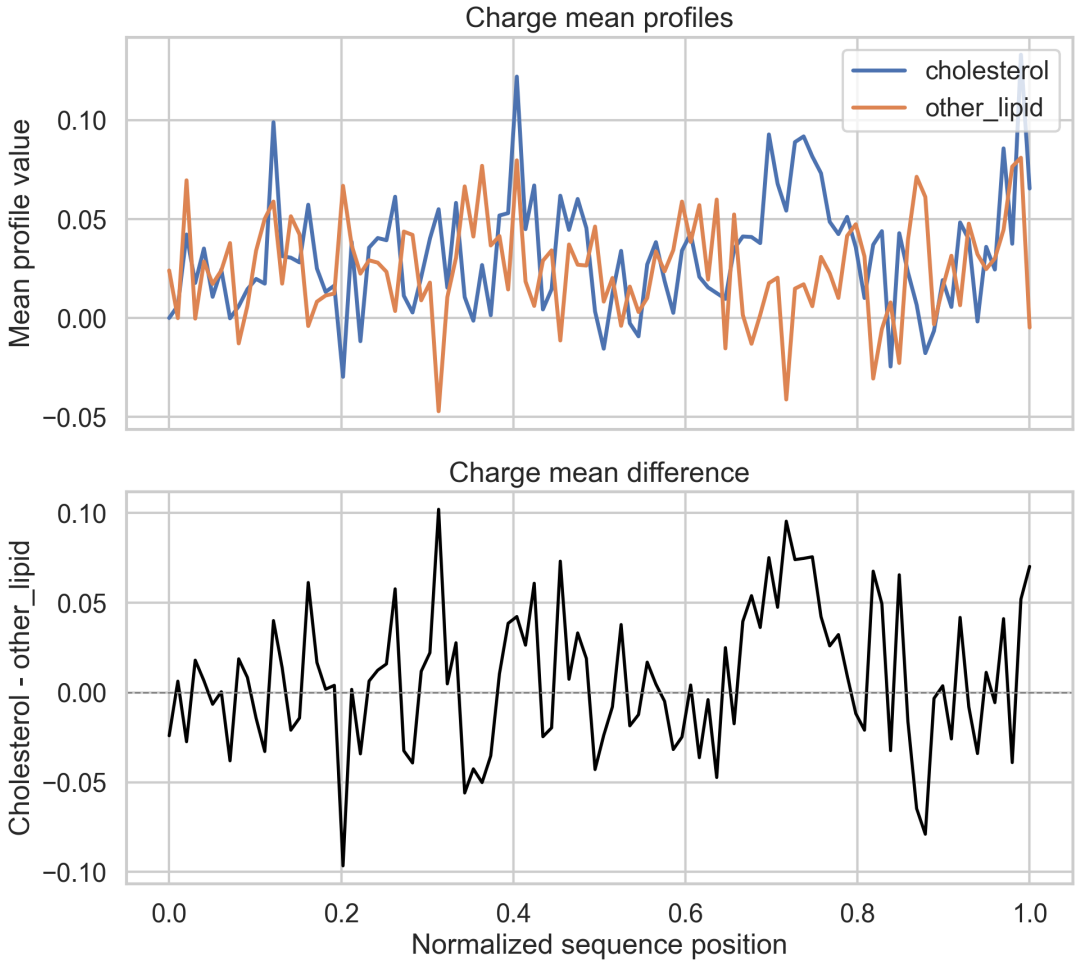

Figure 3: Position-normalized charge profiles for cholesterol-binding and other-lipid-binding proteins. The strongest difference occurred near normalized position ~ 0.31.

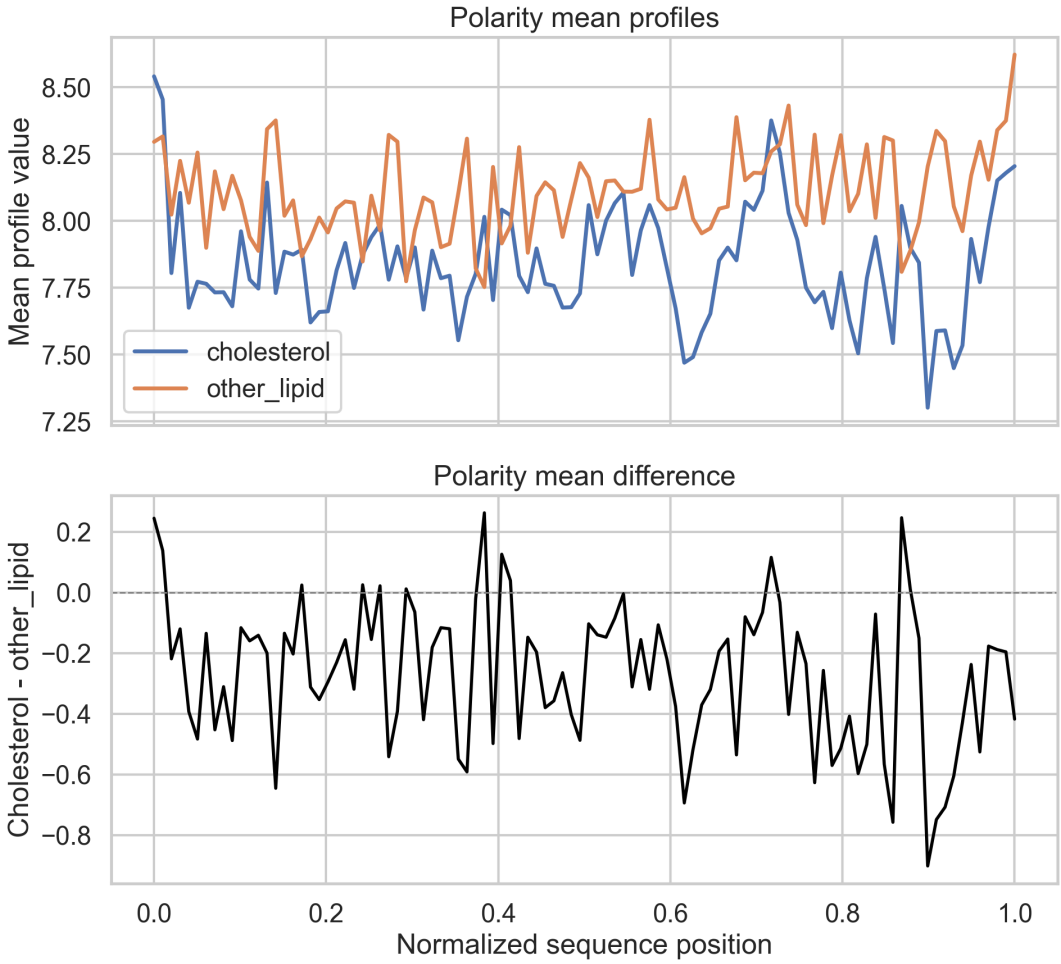

Figure 4: Position-normalized polarity profiles for cholesterol-binding and other-lipid-binding proteins. The largest difference occurred near normalized position ~ 0.90, where the other-lipid class was more polar.

3.5 Classification results as supporting evidence

Classification analyses were performed to assess whether the observed statistical differences translated into predictive signal. These analyses were explicitly treated as supporting evidence rather than the main contribution of the study. Under standard stratified cross-validation, the best-performing model on the homology-reduced dataset was the random forest trained on baseline features, which achieved an accuracy of 0.803, an F1 score of 0.825, and a ROC-AUC of 0.876. A combined random forest using baseline plus profile features performed slightly worse (accuracy 0.772, F1 0.810, ROC-AUC 0.842), and the combined model without hydrophobic features performed intermediately (accuracy 0.793, F1 0.825, ROC-AUC 0.853).

These results reinforce two points. First, interpretable global sequence-composition features already

contain substantial discriminatory signal. Second, explicit hydrophobicity contributes importantly to the classification problem, but the hydrophobic-ablation model retained substantial performance, indicating that hydrophobicity is not the sole source of separability. In contrast, profile-only models were weaker, suggesting that position-dependent profiles provide useful descriptive information but do not outperform simpler baseline composition measures for this task.

Table 4: Representative standard cross-validation classification results. Classification was used as supporting evidence rather than as the primary endpoint of the study.

Model	**Accuracy**	**F1**	**ROC-AUC**
Baseline random forest	0.803	0.825	0.876
Combined without hydrophobic features random forest	0.793	0.825	0.853
Combined random forest	0.772	0.810	0.842

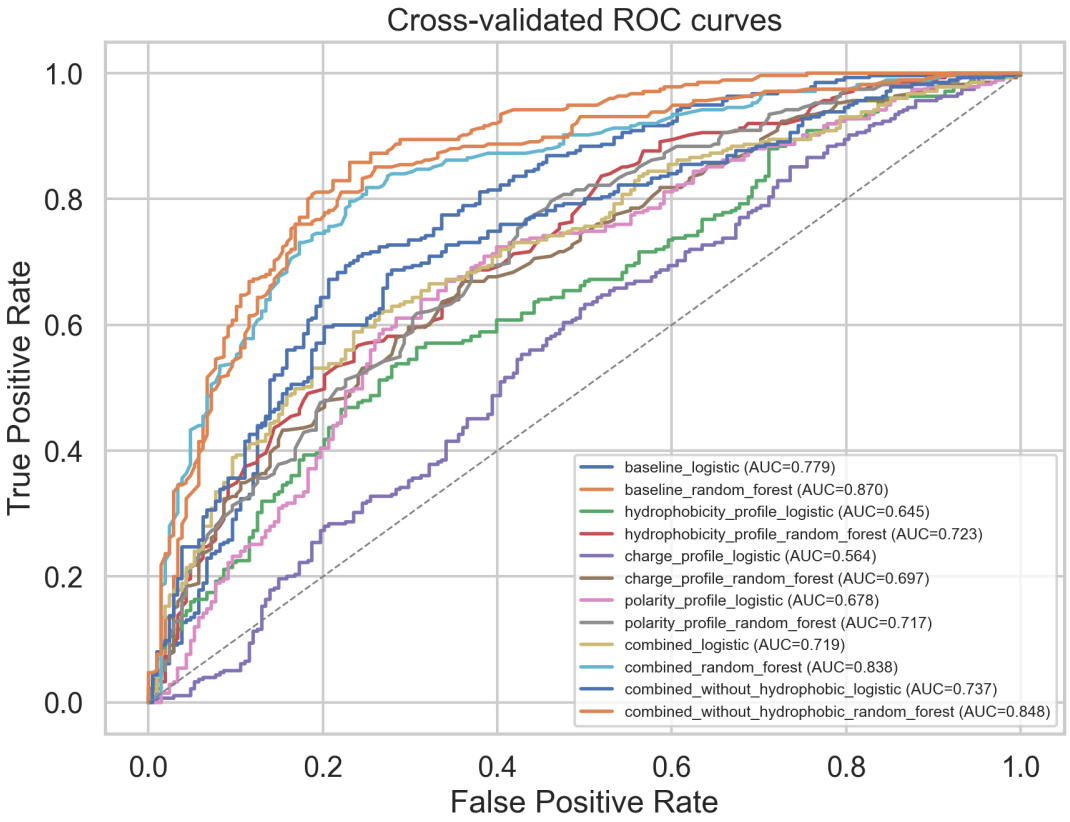

Figure 5: Receiver operating characteristic curves for Stage 7 standard cross-validation analyses. Baseline compositional models outperformed profile-only and combined feature sets.

3.6 Cluster-aware validation reveals residual homology inflation but preserves strong signal

The cluster-aware analysis provided the most important validation of generalization. Under ordinary standard cross-validation on the full cleaned dataset, the baseline random forest achieved an accuracy of 0.919, an F1 score of 0.933, and a ROC-AUC of 0.973. Under cluster-aware cross-validation, where all proteins from the same CD-HIT cluster were confined to the same fold, the same model achieved an accuracy of 0.806, an F1 score of 0.840, and a ROC-AUC of 0.864. Thus, the stricter family-separated evaluation reduced ROC-AUC by 0.109.

The same pattern was observed for the combined random forest, whose ROC-AUC decreased from 0.932 to 0.822, also a drop of approximately 0.110. Logistic regression models also declined under cluster-aware evaluation: baseline logistic regression fell from 0.819 to 0.749 ROC-AUC, and combined logistic regression fell from 0.783 to 0.706. These decreases show clearly that ordinary cross-validation was inflated by residual family similarity.

However, the predictive signal did not disappear when homologous clusters were held out. The cluster-aware baseline random forest retained a ROC-AUC of 0.864, which remains a strong result for a sequence-based and fully interpretable comparison between two biologically related classes. Accordingly, the data support a balanced conclusion: homology inflated standard performance estimates, but robust class-discriminating sequence signal persists even after homology-aware validation.

Table 5: Comparison of standard and cluster-aware validation. Cluster-aware evaluation used homology-cluster groups to prevent residual family leakage between training and test folds.

Model	**Standard CV**			**Cluster-aware CV**		
	Acc.	**F1**	**AUC**	**Acc.**	**F1**	**AUC**
Baseline random forest	0.919	0.933	0.973	0.806	0.840	0.864
Combined random forest	0.869	0.895	0.932	0.784	0.829	0.822
Baseline logistic regression	0.781	0.828	0.819	0.716	0.773	0.749
Combined logistic regression	0.733	0.781	0.783	0.667	0.722	0.706

For the two random-forest models, the ROC-AUC decrease under cluster-aware validation was approximately 0.109–0.110, indicating substantial but incomplete homology-driven inflation in ordinary cross-validation.

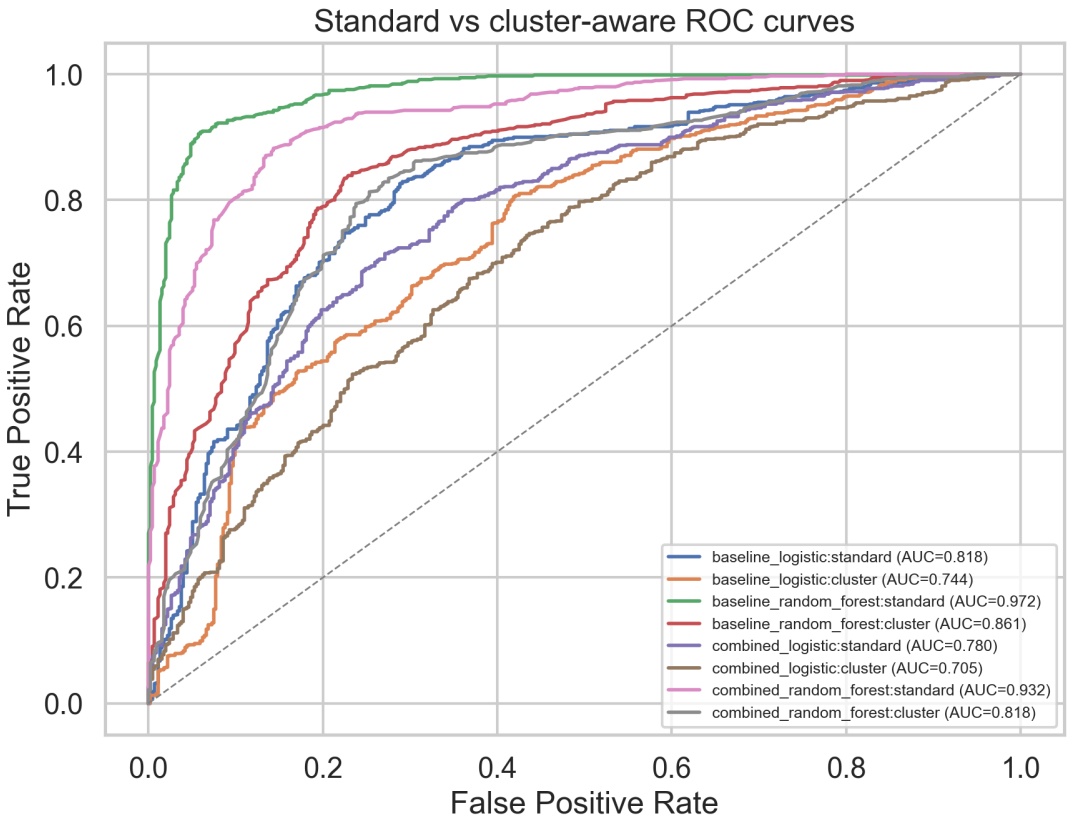

Figure 6: Comparison of standard and cluster-aware validation. Performance declined under cluster-aware evaluation, indicating residual homology inflation in ordinary cross-validation, but substantial predictive signal remained.

3.7 Local motif and local-pattern analysis

The local-pattern follow-up analysis showed that the strongest motif-related signal came from broad local membrane-like organization rather than from simple cholesterol-consensus patterns alone. Cholesterol-binding proteins had substantially higher maximum local hydrophobic window scores (0.950 versus 0.866, FDR 2.86×10^{-38}), longer longest hydrophobic runs (10.94 versus 10.03, FDR 2.78×10^{-36}), and higher mean local hydrophobic window scores (FDR 5.58×10^{-29}). They also more frequently contained aromatic-enriched windows (0.960 versus 0.800, FDR 2.71×10^{-17}), suggesting stronger local membrane-interface-like sequence organization.

Cysteine-cluster features were likewise enriched in the cholesterol-binding class. The presence of `CXXC` patterns was 0.240 versus 0.073 (FDR 8.78×10^{-14}), and `CC` and `CXC` patterns were also more common in cholesterol-binding proteins. These findings are consistent with the earlier global cysteine enrichment and suggest that local cysteine organization may be part of the broader cholesterol-associated sequence context, although the mechanistic basis remains uncertain.

By contrast, simple CRAC- and CARC-like patterns were only modestly informative. `CRAC`-like motifs were very common in both classes, although slightly more prevalent in cholesterol-binding proteins (0.974 versus 0.912, FDR 6.44×10^{-6}). `CARC`-like motifs were nearly ubiquitous in both groups (0.994 versus 0.976, FDR 0.014). Thus, while these motif-like descriptors show some enrichment, they are clearly insufficient as standalone cholesterol-specific signatures in this dataset. The stronger and more biologically plausible local signal lies in broader hydrophobic and aromatic organization rather than in the presence of a short consensus motif alone.

An optional cluster-aware predictive-value test showed that motif/local-pattern features carried substantial signal on their own, but they did not improve upon the baseline feature set. Motif-only random forests achieved a ROC-AUC of 0.850, whereas the baseline-only random forest achieved 0.863 and the baseline-plus-motif random forest achieved 0.861. This suggests that motif/local-pattern features largely recapitulate the same membrane-adaptation signal already captured by the baseline features rather than providing a large new orthogonal source of discrimination.

Table 6: Selected motif and local-pattern differences between cholesterol-binding and other-lipid-binding proteins.

Feature	**Cholesterol**	**Other lipid**	**FDR**
`max_local_hydrophobic_window_score`	0.950	0.866	2.86×10^{-38}
`longest_hydrophobic_run`	10.94	10.03	2.78×10^{-36}
`mean_local_hydrophobic_window_score`	0.537	0.491	5.58×10^{-29}
`aromatic_enriched_window_present`	0.960	0.800	2.71×10^{-17}
`cxxc_present`	0.240	0.073	8.78×10^{-14}
`crac_present`	0.974	0.912	6.44×10^{-6}
`carc_present`	0.994	0.976	0.014

3.8 Results summary

Taken together, the results support a coherent view of cholesterol specificity at the sequence level. Cholesterol-binding proteins differ from other lipid-binding proteins in global hydrophobicity, polarity, acidity, and selected residue enrichments; they also differ in the position-dependent organization of hydrophobicity, charge, and polarity. Predictive models built from these features perform well, although performance declines under cluster-aware validation, confirming that homology contributes to but does not fully explain the signal. Local-pattern analyses further indicate that cholesterol-binding proteins are enriched in broader hydrophobic and aromatic membrane-like segments and in selected cysteine-cluster patterns, whereas simple CRAC- or CARC-like motifs alone are not sufficient to account for the observed class differences.

4 Discussion

This study was designed to address a narrow but biologically important question: are cholesterol-binding proteins distinguishable from other lipid-binding proteins at the sequence level, and if so, what features drive that distinction? The results support an affirmative answer. Across multiple layers of analysis, cholesterol-binding proteins consistently appeared more hydrophobic, less polar, less acidic, and enriched for selected membrane-associated residue types and local sequence organizations. These differences persisted after rigorous cleaning, removal of proteins observed in both classes, homology reduction, and cluster-aware validation.

The biological interpretation that emerges is not one of a single cholesterol-binding sequence motif that is universally present across proteins. Instead, cholesterol-binding proteins appear more strongly adapted to membrane-embedded or membrane-interface environments in a distributed manner. The enrichment of hydrophobic and aromatic residues, longer hydrophobic runs, stronger local hydrophobic windows, and lower overall polarity is consistent with proteins that either contain more transmembrane sequence, present larger interfacial surfaces compatible with sterol interaction, or require more strongly membrane-adapted sequence architecture to maintain cholesterol-sensitive conformational states. This interpretation fits well with the known physicochemical role of cholesterol as a rigid, interfacially anchored sterol whose effects are closely tied to membrane order, packing, and protein–bilayer coupling.

One notable observation is that the simplest global composition features outperformed profile-only and combined feature sets in classification. This may seem counterintuitive at first, because profile features carry additional positional information. However, several factors make the result plausible. First, the strongest discriminatory signal may indeed be largely global, reflecting overall membrane adaptation rather than fine-grained sequence topology. Second, position-normalized interpolation compresses proteins of widely varying architecture into a common coordinate system, which is useful descriptively but may blur family-specific spatial organization when used directly for prediction. Third, adding high-dimensional profile features to a moderate-sized dataset can dilute robust low-dimensional signal, especially when the dominant differences are already captured by baseline physicochemical summaries. In that sense, the classification results strengthen rather than weaken the argument for interpretable baseline features.

The local-pattern analysis adds an important nuance to this picture. Although the strongest local signals again reflected broad hydrophobic and aromatic organization rather than short motifs, these local features still provide evidence that the cholesterol-associated sequence pattern is not purely global. Cholesterol-binding proteins were enriched for aromatic-enriched windows and showed stronger local hydrophobic segments, suggesting that local sequence patches compatible with membrane-interface stabilization may contribute to cholesterol sensitivity. This is biologically plausible, because aromatic residues frequently occupy membrane interfacial positions and can help stabilize regions where cholesterol headgroup and ring structure encounter the protein surface.

The cysteine-cluster signal deserves particular attention. Cysteine enrichment emerged both in global amino-acid composition and in local `CC`, `CXC`, and `CXXC` patterns. Several interpretations are possible. Cysteine-rich local organization may reflect structural stabilization, conserved disulfide-forming extracellular motifs in some membrane receptor classes, family composition biases, or localizable microenvironments associated with cholesterol-sensitive proteins. The present study cannot resolve these possibilities mechanistically, but it does highlight cysteine-associated local architecture as a feature worthy of deeper follow-up.

By contrast, the CRAC/CARC motif analysis produced a more restrained conclusion. CRAC- and CARC-like patterns were somewhat enriched in cholesterol-binding proteins, but they were also extremely common in the other-lipid class, especially once the comparison was restricted to membrane-associated lipid-binding proteins. This indicates that these motifs are not sufficient as standalone cholesterol-specific signatures in a realistic sequence-comparison setting. Their prevalence likely reflects the fact that short, permissive patterns involving hydrophobic residues, aromatic residues, and basic residues can arise frequently in membrane proteins. Accordingly, our findings support a cautious interpretation of CRAC/CARC scans: they are reasonable heuristic descriptors, but they should not be treated as definitive evidence of direct cholesterol recognition without structural or functional corroboration.

The cluster-aware validation results are central to the credibility of the study. The marked drop in performance from standard to cluster-aware cross-validation demonstrates that family-level similarity inflated ordinary cross-validation estimates. This was expected and, importantly, explicitly tested rather than ignored. At the same time, substantial predictive signal remained under the stricter evaluation, with the baseline random forest maintaining a cluster-aware ROC-AUC of 0.864. This is a key result because it shows that the sequence differences identified here are not merely artifacts of redundant homologs. Instead, they likely reflect broader biochemical tendencies that generalize across at least part of the homology structure of the dataset.

4.1 Limitations

Several limitations should be considered. First, ligand annotation remains an inherent limitation of any structure-derived interaction study. Even curated resources such as BioLiP depend on the underlying structural record, and the presence of a ligand in a structure does not by itself prove physiological specificity. Some proteins may bind cholesterol opportunistically in experimental settings, and some relevant lipid interactions may be absent because they were not structurally captured.

Second, residual family composition effects may still influence the results even after clustering and cluster-aware validation. The dataset is not family-balanced, and some protein families are likely overrepresented in the cholesterol class or in the comparison class. Cluster-aware validation reduces leakage, but it does not create a taxonomically or family-matched experimental design.

Third, the present study is sequence-only in scope. Sequence-level differences are valuable and interpretable, but they cannot substitute for explicit structural analysis of binding pockets, membrane insertion geometry, residue orientation, or lipid-facing surfaces. A future extension should therefore integrate structure-based descriptors, especially given the known importance of local three-dimensional context in sterol binding.

Fourth, the motif and local-pattern analysis remains heuristic. CRAC-like, CARC-like, and cysteine-cluster rules are useful descriptive features, but they do not establish mechanistic motifs in the strict structural sense. Enrichment of such patterns may reflect broad membrane-protein organization rather than direct ligand-contact determinants.

Finally, the entire comparison may depend partly on membrane-protein enrichment. Cholesterol-binding proteins are expected to be disproportionately membrane-associated, and although the comparison class was restricted to other lipid-binding proteins rather than arbitrary non-binders, differences in membrane topology or domain composition could still contribute to the observed separation.

4.2 Future directions

Several next steps follow naturally from this work. Structure-based analysis could test whether the sequence signals identified here correspond to specific classes of binding-site geometry or sterol-facing residues. Family-aware analyses could examine whether the same trends hold within GPCRs, transporters, channels, and enzymes separately. Motif-context analyses could determine whether CRAC-like or cysteine-cluster patterns become more informative when restricted to predicted transmembrane or membrane-proximal segments. Finally, matched designs controlling for taxonomy, membrane topology, or family membership could help refine the specificity question further.

Overall, the present study establishes a rigorous sequence-level baseline and suggests that cholesterol specificity is best understood as a distributed biochemical and architectural signal rather than as a single universal motif.

5 Conclusion

We developed a reproducible and Overleaf-ready sequence-analysis framework to compare cholesterol-binding proteins with proteins that bind other biologically meaningful lipids. By combining careful ligand curation, protein-level cleaning, exact-sequence deduplication, removal of multi-lipid proteins, homology reduction, interpretable feature engineering, and cluster-aware validation, we showed that cholesterol-binding proteins are distinguishable from other lipid-binding proteins by coherent sequence-level properties.

The strongest signals were not limited to short motif-like patterns. Instead, cholesterol-binding

proteins were characterized by higher hydrophobicity, lower polarity, lower acidic residue content, enrichment of selected residues including cysteine, leucine, and phenylalanine, stronger local hydrophobic and aromatic membrane-like segments, and persistent discriminative signal under family-separated evaluation. At the same time, the study showed that standard cross-validation can materially overestimate performance in this setting, underscoring the importance of homology-aware validation for biologically credible claims.

Taken together, these results support a model in which cholesterol specificity is reflected by distributed membrane-adapted sequence architecture rather than by a single universally diagnostic motif. This manuscript therefore provides both a substantive biological conclusion and a rigorous computational template for future comparative studies of lipid-specific protein recognition.

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